1 Coordinates



Definition 1.1 (Ordered Basis)

Let V be a finite-dimensional vector space. An ordered basis of V is a basis $B = \{v_1, v_2, ..., v_n\}$ together with a chosen ordering of the basis vectors. We denote an ordered basis as an n-tuple $(v_1, v_2, ..., v_n)$.

Let V be a finite-dimensional vector space over F and $B = (v_1, v_2, ..., v_n)$ an ordered basis of V. We proved last time that the linear combination map $L_B : F^n \to V$ defined by

$$L_B(x_1, x_2, ..., x_n) = x_1v_1 + x_2v_2 + ... + x_nv_n,$$

is an isomorphism. We will call this the basis isomorphism with respect to B. The inverse isomorphism $L_B^{-1} : V \to F^n$ takes a vector v inV to the point $(x_1, ..., x_n) \in F^n$, which are the coefficients of v with respect to the basis B. We call L_B^{-1} the coordinate isomorphism with respect to B. We call $[v]_B := (x_1, ..., x_n) \in F^n$ the *coordinate vector of v with respect to B*. When B is understood, we write $[v]$ for the coordinate vector of V.



Remark 1.2

Note that the use of an ordered basis here allows us to specify a first coordinate, second coordinate, etc., of $[v]$.



Example 1.3

Let $V = F^n$ and $B = (e_1, e_2, ..., e_n)$, where e_i is the n-tuple with a 1 in the ith position and 0s elsewhere. Then for $x = (x_1, x_2, ..., x_n) \in V$,

$$L_B(x) = L_B(x_1, ..., x_n) = x_1e_1 + ... + x_ne_n = (x_1, ..., x_n) = x,$$

showing $L_b = \mathrm{id}_{F^n}$ (in which case we have $L_B^{-1} = \mathrm{id}_{F^n}$).



Example 1.4

Let $V = \mathcal{P}_n(F)$ and $B = (1, x, x^2, ..., x^n)$. Then for $p = a_0 + a_1x + ... + a_nx^n \in \mathcal{P}_n(F)$,

$$[p]_B = L_B^{-1}(a_0 + a_1x + ... + a_nx^n) = (a_0 + a_1x + ... + a_nx^n) = (a_0, a_1, ..., a_{n+1}) \in F^{n+1}.$$



Remark 1.5

Since $F^n \cong M^{n\times 1}(F)$, we often write $(x_1, ..., x_n)$ interchangeably with $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ without comment.

2 Matrix of a Linear Transformation

We now show that every linear map $T : V \to W$ between finite-dimensional vector spaces can be represented by a matrix. Choose ordered bases $B_1 = (v_1, ..., v_n)$ and $B_2 = (w_1, ..., w_m)$ of V and W, respectively. Since $Tv_j \in W$, there exist unique scalars T_{ij} such that

$$Tv_j = \sum_{i=1}^{m} T_{ij}w_i. \tag{1}$$



Definition 2.1 (Matrix of a Linear Map)

We define *the matrix of T with respect to bases B_1 and B_2* denoted by $[T]_{B_2,B_1}$, to be the $m \times n$ matrix with entries T_{ij} defined by (2.1). From (2.1), we see that for $j = 1, ..., n$, the jth column is $[Tv_j]_{B_2}$, the coordinate vector of Tv_j with respect to B_2. When B_1 and B_2 are understood, we write $[T]$ for the matrix of T.



Example 2.2

Let $V = \mathcal{P}_1(F)$, $W = \mathcal{P}_3(F)$, and $T : \mathcal{P}_1(F) \to \mathcal{P}_3(F)$ be defined by $(Tp)(x) = x^2p(x)$. Choose $B_1 = (v_1, v_2) = (1, x)$ and $B_2 = (w_1, w_2, w_3, w_4) = (1, x, x^2, x^3)$. We have:

$$Tv_1 = x^2 \cdot 1 = x^2 = w_3, Tv_2 = x^2 \cdot x = x^3 = w^4,$$

and therefore;

$$[Tv_1]_{B_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, [Tv_2]_{B_1} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Since $[Tv_i]_{B_2}$ is the ith column of the matrix of T,

$$[T]_{B_2,B_1} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

We will now show that applying a linear map T to a vector v is the same as multiplying the coordinate vector of v by the matrix of T. If $v \in V$, we can write $v = \sum_{j=1}^{n}([v])_j v_j$. Then:

$$\begin{aligned} Tv &= T\left(\sum_{j=1}^{n}([v])_j v_j\right) \\ &= \sum_{j=1}^{n}([v]_j)Tv_j \\ &= \sum_{j=1}^{n}([v])_j \sum_{i=1}^{m} T_{ij}w_i \\ &= \sum_{i=1}^{m}\left(\sum_{j=1}^{n} T_{ij}([v])_j\right) w_i, \end{aligned}$$

so:

$$([Tv])_i = \sum_{j=1}^{n} T_{ij}([v])_j, \tag{2}$$

in other words, (2.4) says that the coordinate vector of Tv is given by multiplying the coordinate vector of v by the matrix of T: $[Tv] = [T][v]$.



Example 2.3

Consider again $T \in \mathcal{L}(\mathcal{P}_1(x), \mathcal{P}_3(x))$ defined by $(Tp)(x) = x^2p(x)$. If $p(x) = a_0 + a_1x$, then

$(Tp)(x) = a_0x^2 + a_1x^3$. In Example 2.4 we found the matrix of T is given by

$$[T] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1. \end{bmatrix}$$

The coordinate vector of $p(x0 = a_0 + a_1x$ is $[p] = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$. Multiplying $[T][p]$ gives

$$\left\{ \begin{matrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{matrix} \right\} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ a_0 \\ a_1 \end{bmatrix},$$

which is the coordinate vector of $x^2p(x) = 0 + 0x + a_0x^2 + a_1x^3$.

Exercise 2.4

The matrix of a linearm ap $T \in \mathcal{L}(F^n, F^m)$ with respect to the standard bases of F^n and F^m is called the standard matrix of T. Find the standard matrix of $T \in \mathcal{L}(F^2, F^3)$, where T is defined by:

$$T(x, y) = (x + 3y, 2x + 5y, 7x + 9y).$$

Proof:

We see $(1, 0) \to (1, 2, 7)$ and $(0, 1) \to (3, 5, 9)$, so we get:

$$\begin{bmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{bmatrix}.$$



Example 2.5

Let $D \in \mathcal{L}(\mathcal{P}_3(\mathbf{R}), \mathcal{P}_2(\mathbf{R}))$ be the differentiation map defined by $Dp = p'$. Find the standard matrix of D, and check that $[p'] = [D][p]$ when $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$.

Proof:

We see $x^n \to nx^{n-1}$, so for e_i with $i \neq 3$, $e_i \to ie_{i-1}$, so the matrix becomes:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

and indeed:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1a_1 \\ 2a_2 \\ 3a_3 \\ 0 \end{bmatrix} = 3a_3x^2 + 2a_2x + a_1,$$

as desired.

3 Dimension of $\mathcal{L}(V, W)$

We proved that $\mathcal{L}(V, W)$ is a vector space with respect to the operations

$$(T_1 + T_2)(v) := T_1 v + T_2 V$$
$$(\alpha T)(v) = \alpha T v,$$

we will now prove that when V and W are finite-dimensional with $\dim V = n$ and $\dim W = m$, $\mathcal{L}(V, W)$ is finite-dimensional with $\dim \mathcal{L}(V, W) = mn$.

Exercise 3.1

Fix ordered pairs for V and W an consider the map $\Psi : \mathcal{V}, \mathcal{W} \to M^{m\times n}(F)$ defined by $\Psi(T) \to [T]$. That is, Ψ takes a linear map to its matrix $[T]$ with respect to these two ordered bases. We will prove that Ψ is an isomorphism. Since two vector spaces are isomorphic if and only if they have the same dimension, it follows that $\dim \mathcal{L}(V, W) = \dim M^{m\times n}(F) = mn$.

(a) Prove that Ψ is linear.

(b) Prove that Ψ is injective.

(c) Prove that Ψ is also surjective, and hence an isomorphism.

Proof:

1. We know $\Psi(\alpha T_1 + T_2)$ defines a matrix $[\alpha T_1 + T_2]$ by defining scalars T_{ij} by $(\alpha T_1 + T_2)v_j = \sum_{i=1}^{m} (\alpha T_1 + T_2)_{ij} v_j$. But, by linearity of operators, we have $(\alpha T_1 + T_2)v_j = T_1 \alpha v_j + T_2 v_j = \sum_{i=1}^{m} \alpha T_{1,ij} v_j + \sum_{i=1}^{m} T_{2,ij} v_j$, so equating coefficients (which we can do since the v_j are a basis):

   $$(\alpha T_{1.ij} + T_{2,ij}) = (\alpha T_1 + T_2)_{ij},$$

   showing linearity in each entry of the matrix and therefore in the matrices themselves, so we have:

   $$[\alpha T_1 + T_2] = \alpha [T_1] + T_2,$$

   showing linearity of Ψ.

2. If $\Psi(T) = [T] = 0 \in M^{m\times n}(F)$, then every entry T_{ij} is zero. This implies;

   $$T v_j = \sum_{i=1}^{m} T_{ij} v_j = 0$$

   for all v_j, and therefore $Tv = 0$ for all $v \in V$ since the v_j form a basis. Therefore, T is the zero vector and therefore the kernel is trivial, showing Ψ is injective.

3. Given a matrix $[T]$ with entries T_{ij}, define a linear map T on the basis by $T v_j = \sum_{i=1}^{m} T_{ij} v_j$. We can check this is linear on the basis, and:

   $$T(\alpha v_j + v_k) = \sum_{i=1}^{m} T_{ij} (\alpha v_j + v_k) = \alpha \sum_{i=1}^{m} T_{ij} v_j + \sum_{i=1}^{m} T_{ij} v_k = (v_j) + T(v_k),$$

   showing T is linear. By construction, $\Psi(T) = [T]$, so Ψ is surjective.

4 Product of Linear Maps

We previously defined a product of linear operators in $\mathcal{L}(V)$ by composition: $T_2T_1 = T_2 \circ T_1$.

Exercise 4.1

Assume V is finite-dimensional and let $(v_1, ..., v_n)$ be an ordered basis of V.

(a) Show that $[\mathrm{id}_V] = I_n$, the $n \times n$ identity matrix.

(b) Show that $[T_2T_1] = [T_2][T_1]$. That is, the matrix representing the linear operator T_2T_1 is the product of the matrices representing the linear operators T_2 and T_1.

In other words, the vector space isomorphism $\mathcal{L}(V) \to M^{m\times n}(F)$ from the previous section is actually an isomorphism of unital associative F-algebras.

Proof:

(a) We have $\mathrm{id}_V(v_j) = \sum_{i=0}^{m} \alpha_{ij}v_j$ by $\alpha_{ij} = 0$ unless $i = j$ (simply by duh), so $[\mathrm{id}_V]_{ij} = \delta_{ij}$, and we recover the identity.

(b) Let V be m-dimensional, W be n-dimensional, U be q-dimensional, and $T_1 : V \to W$, $T_2 : W \to V$. Then, $[T_2][T_1]_{ij} = \sum_{l=1}^{n} T_{2,il}T_{1,lj}$. We have:

$$\begin{aligned}(T_2T_1)(v_j) &= \sum_{i=1}^{q}[T_2T_1]_{ij}v_j \\ &= \sum_{i=1}^{q}\left(\sum_{l=1}^{n} T_{2,il}T_{1,lj}\right)v_j \\ &= \sum_{l=1}^{q} T_{2,il}\left(\sum_{i=1}^{m} T_{1,lj}\right)v_j = \sum_{i=1}^{q} T_{2,ij}T(v_j)...\end{aligned}$$

Exercise 4.2

Let $V = \mathcal{P}_2(\mathbf{R})$ with the standard ordered basis. Consider $T \in \mathcal{L}(V)$ defined by $(Tp)(x) = xp'(x)$.

Proof:

This is the map of multiplication sending x^i to ix^i, so we have:

$$[T] = \begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 0 & 3,\end{bmatrix},$$

so:

$$[T]^2 = \begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 4 & 0\\ 0 & 0 & 0 & 9,\end{bmatrix},$$

T^2 is the map sending $x^i \to ix^i \to i^2x^i$, so this has matrix:

$$[T^2] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 9, \end{bmatrix},$$

as desired.

